

POWER CONSIDERATIONS

The average chip-junction temperature, T_J , in $^{\circ}\text{C}$ can be obtained from:

$$T_J = T_A + (P_D \cdot \theta_{JA}) \quad (1)$$

where:

T_A = Ambient Temperature, $^{\circ}\text{C}$

θ_{JA} = Package Thermal Resistance,
Junction-to-Ambient, $^{\circ}\text{C}/\text{W}$

P_D = $P_{\text{INT}} + P_{\text{I/O}}$

P_{INT} = $I_{\text{CC}} \times V_{\text{CC}}$, Watts — Chip Internal Power

$P_{\text{I/O}}$ = Power Dissipation on Input and Output Pins
— User Determined

For most applications $P_{\text{I/O}} < P_{\text{INT}}$ and can be neglected.

The following is an approximate relationship between P_D and T_J (if $P_{\text{I/O}}$ is neglected):

$$P_D = K \div (T_J + 273^{\circ}\text{C}) \quad (2)$$

Solving equations (1) and (2) for K gives:

$$K = P_D \cdot (T_A + 273^{\circ}\text{C}) + \theta_{JA} \cdot P_D^2 \quad (3)$$

where K is a constant pertaining to the particular part. K can be determined from equation (3) by measuring P_D (at equilibrium) for a known T_A . Using this value of K , the values of P_D and T_J can be obtained by solving equations (1) and (2) iteratively for any value of T_A .

The total thermal resistance of a package (θ_{JA}) can be separated into two components, θ_{JC} and θ_{CA} , representing the barrier to heat flow from the semiconductor junction to the package (case) surface (θ_{JC}) and from the case to the outside ambient (θ_{CA}). These terms are related by the equation:

$$\theta_{JA} = \theta_{JC} + \theta_{CA} \quad (4)$$

θ_{JC} is device related and cannot be influenced by the user. However, θ_{CA} is user dependent and can be minimized by such thermal management techniques as heat sinks, ambient air cooling, and thermal convection. Thus, good thermal management on the part of the user can significantly reduce θ_{CA} so that θ_{JA} approximately equals θ_{JC} . Substitution of θ_{JC} for θ_{JA} in equation (1) will result in a lower semiconductor junction temperature.

Values for thermal resistance presented in this document, unless estimated, were derived using the procedure described in Motorola Reliability Report 7843, "Thermal Resistance Measurement Method for MC68XX Microcomponent Devices," and are provided for design purposes only. Thermal measurements are complex and dependent on procedure and setup. User derived values for thermal resistance may differ.